Attendance Management System Using Face Recognition

[1]Rohini Vhatkar, [2]Anas Shaikh, [3]Qadri Hasan, [4]Prof. Dinesh Deore

[1,2,3]Student, [4]Professor
Department of Electronics and Telecommunication, Rizvi College of Engineering Mumbai, India

Abstract: This project involves building an attendance system which utilizes facial recognition to mark the presence, time-in, and time-out of employees. It covers areas such as facial detection, alignment, and recognition, along with the development of a web application to cater to various use cases of the system such as registration of new employees, addition of photos to the training dataset, viewing attendance reports, etc. This project intends to serve as an efficient substitute for traditional manual attendance systems. It can be used in corporate offices, schools, and organizations where security is essential. A Face recognition system is an application of computer vision and image processing which is capable of performing two major tasks of identifying and verifying a person from an image or a video database. The main objective is to automate the attendance system by integrating the face recognition technology using modified Local Binary Pattern (LBP) and Support Vector Machine (SVM) algorithm with Matlab Graphical User Interface (GUI). Once face detection and feature extraction is performed by using viola jones and LBP, then the image is further moved for recognition. If the image is recognized then the attendance of student is marked as present in the data base, else it marked as absent.

Key words: Attendance system, Python, Machine Learning, HTML, MySQL.

INTRODUCTION:

The Attendance System using Face – Recognition is a replacement way method for the traditional way of marking attendance. The proposed system is python, tkinter based system supported with MySQL database. System of a particular institute. This system is proposed to be based on biometrics .i.e. Face Authentication. Since there is presence of biometrics, this system completely eliminates the chances of fake attendance which is a problem with the traditional methods of attendance. The Attendance management is the significant process that were carry out in every institute to monitor the performance of the student. Every institute does this is its own way. For performing encoding Deep Conversional Neural Network algorithm will get carried out & for each face 128 measurements were generated then the measurements of the face that were detected in frame it get compared with the measurements of the faces that were present in the image which is earlier stored in the database. So at last by using simple liner SVM algorithm system will find the person in database of know peoples (i.e. capture at the starting of the project) who has closest measurements to the image that were detected by camera. After finding perfect match system will generate the name and date & time & present mark and store the entry in CSV file. Which were further uploaded on the database and also user can open it with Microsoft Excel.

LITERATURE SURVEY

Using real time computer vision algorithms in automatic attendance management systems this paper introduces a new approach in automatic attendance management systems, extended with computer vision algorithms. The Proposed system uses real time face detection algorithms integrated on an existing Learning Management System (LMS), which automatically detects and registers students attending on a lecture. The system represents a supplemental tool for instructors, combining algorithms used in machine learning with adaptive methods used to track facial changes during a longer period of time.

Face Recognition-based Lecture Attendance System proposed a system that takes the attendance of students for classroom lecture. The system takes attendance automatically using face recognition. However, it is difficult to estimate the attendance precisely using each result of face recognition independently because the face detection rate is not sufficiently high. In this paper, we propose a method for estimating the attendance precisely using all the results of face recognition obtained by continuous observation.

Automatic Control of students' attendance in Classrooms Using RFID Radio frequency identification (RFID) is one of the automatic identification technologies more in vogue nowadays. There is a wide research and development in this area trying to take maximum advantage of this technology, and in coming years many new applications and research areas will continue to appear.

Attendance management system using hybrid face recognition techniques [2] which uses modified viola-jones algorithm and free partial face recognition algorithm for face recognition for face detection for high accuracy. The method uses Gabor Ternary Pattern (GTP) for robust and discriminative face recognition. This system improves the existing attendance recording system by avoiding manual recording of attendance. The efficiency of face recognition depends upon the number of faces detected. The higher the detected faces the higher the face recognition rate. The percentage of detected faces is 95% and percentage of recognized faces is 47.36% and it keeps changing for each frame.

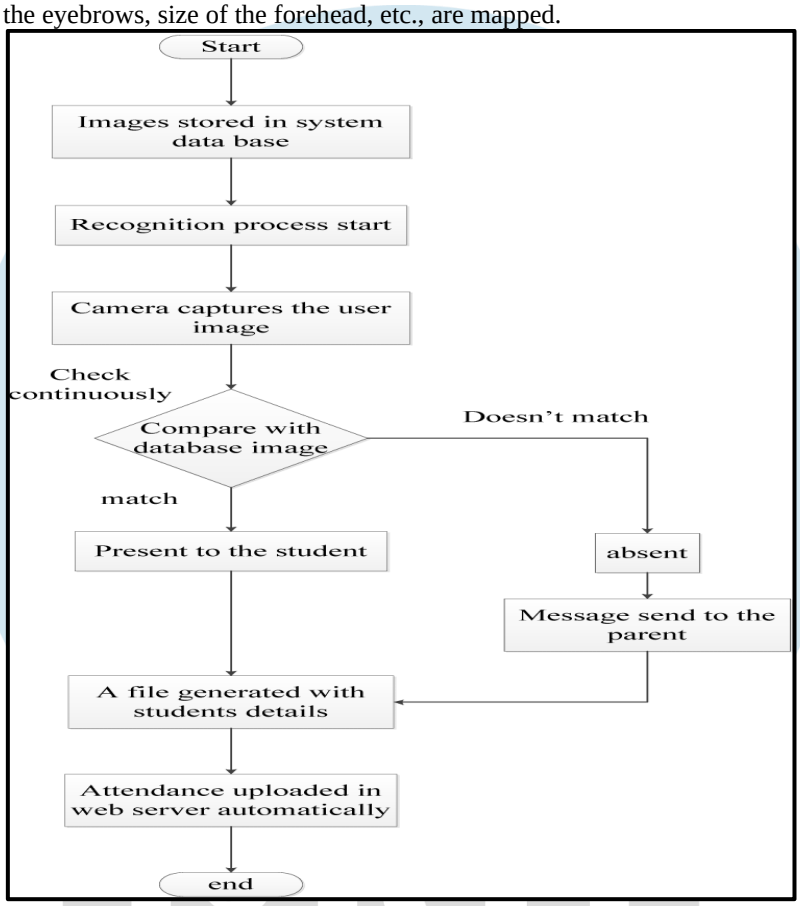
Face Recognition based Attendance Management System using Machine Learning the most arduous task in any organization is attendance marking. In this paper we proposed an automated attendance management system which tackles the predicament of recognition of faces in biometric systems subject to different real time scenarios such as illumination, rotation and scaling.

The use of asymmetric local binary pattern for facial expression recognition along with the modified convolution techniques [4]. The use of AS-LBP results in high rate of face recognition without the loss of appearance information. The face image is detected with the help of viola and jones face and eye detectors. Support vector machine is used to identify the facial expression which has high accuracy. The face recognition rate increases when the width is greater than the height of the operator and results in maximum of 95.71% for 13x3 and 15x3 sizes with an exception for 5x7 size. When the width and height of the operator is equal (observe diagonally from top left to right bottom) the face recognition rate will not increase, though there is an increase in the operator and

79.46% on FGNET database for the size 15x7. It is inequitable to compare with same work performed using LBP as the face recognition rate is dependent on some parameters such as image size, image sub-region size, databases used, classifier parameters. Face Detection and Recognition Using Skin Color [5]. The performance of face detection, which uses skin color division with thresholding skin color model combined with AdaBoost algorithm. The facial properties extraction is obtained with the help of Principal Component Analysis (PCA) and K-Nearest Neighbor (KNN) based classification. Additionally, they have used Morphological Operators to improve the face detection performance. Some images for which the expected results have not been reached are no proper orientation of faces, image illumination is not good and if the distance of faces from the camera is different. For very few images that vary in the high level of orientation are not detected and recognized properly. This gives around 96% recognition rate for the available database.

SYSTEM DESIGN

Face recognition works by inputting he image into the database before being ready to use. Once the image is processed, it converts it into an analogues and gets saved in the database. Face analysis is a technique in which facial features like the length of the nose, lips, size between the eyebrows, size of the forehead, etc., are mapped.



Block diagram for Attendance Management system

These features are recorded and saved in the system. Once the database preserves the image, it is detected by the camera. To use it successfully, the system be used correctly. The system must be able to identify the individuals being analyzed. It should be used with individuals who are known, in advance, to be in database.

A. Face detection

Hear classifier based method is chosen for face detection owing to its high detection accuracy and real time performance. Hear like features encodes the difference in average intensity in the different parts of the image and consists of black and white connected rectangles in which the value of the feature is the difference of sum of pixel values in black and white regions. Hear like features can be found out at different scales and positions which are robust for detection purpose. The computational speed of the feature calculation is increased with the use of Integral image. The number of Haar features available from a small image is very large. The optimal set of features to be used and their corresponding thresholds to be used for classification is obtained from AdaBoost algorithm.

B. Feature Extraction

Selection of the feature vector is the most important part in a pattern classification problem. The inherent problems related to image classification include the scale, pose, translation and variations in illumination level. The feature vector extracted should be invariant to these problems to achieve better accuracy in classification. LBP based feature extraction method is used owing to its excellent light invariance property and low computational complexity. The neighborhood values are threshold by the center value and the result is treated as a binary number. In this way, it encodes the neighborhood information very efficiently.

C. Classification

To classify the feature obtained from the LBP technique, multi class Support Vector Machines (SVMs) are used. SVM have developed from Statistical Learning Theory. They have been widely applied to fields such as character, handwriting digit and text recognition, and more recently to satellite image classification. SVMs, like ASVM and other nonparametric classifiers have a

reputation for being robust. SVMs function by nonlinearly projecting the training data in the input space to a feature space of higher dimension by use of a kernel function.

IV.Testing

Log in to the system.

Case 1: Invalid Username or password entered by the user.
Output: Error message on the screen saying "Invalid credentials"
Case 2: Valid credentials.
Output: The user is redirected to the Dashboard page.

Update Profile.

Case 1: Username already exists.
Output: Error message on the screen saying "Username already exists"
Case 2: Some of required fields missing in input.
Output: Model validation errors will be displayed to the user.
Case 3: All input data are valid.
Output: Profile updated successfully.

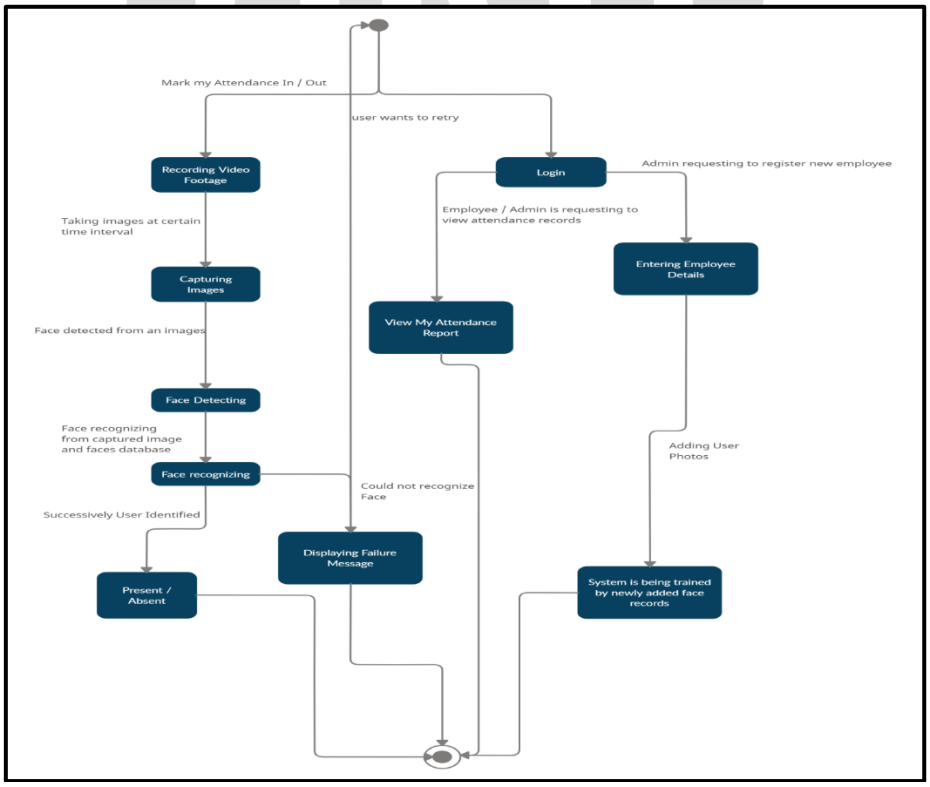
View Attendance.

Case 1: User is not logged in.
Output: Redirected to the login page with error message "Please login!".
Case 2: If a user exists and has the attendance records
Output: All the chat history will be displayed
Case 3: Provided username does not exist in the system.
Output: 404 Error

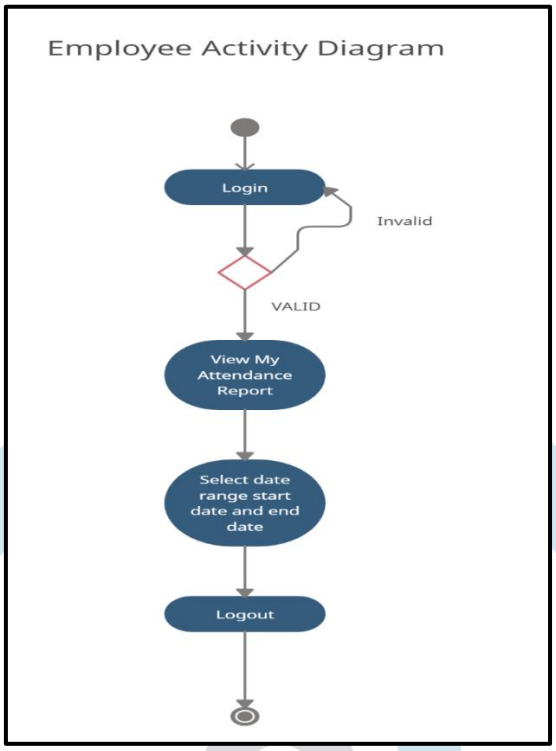
V.Basic System Functioning

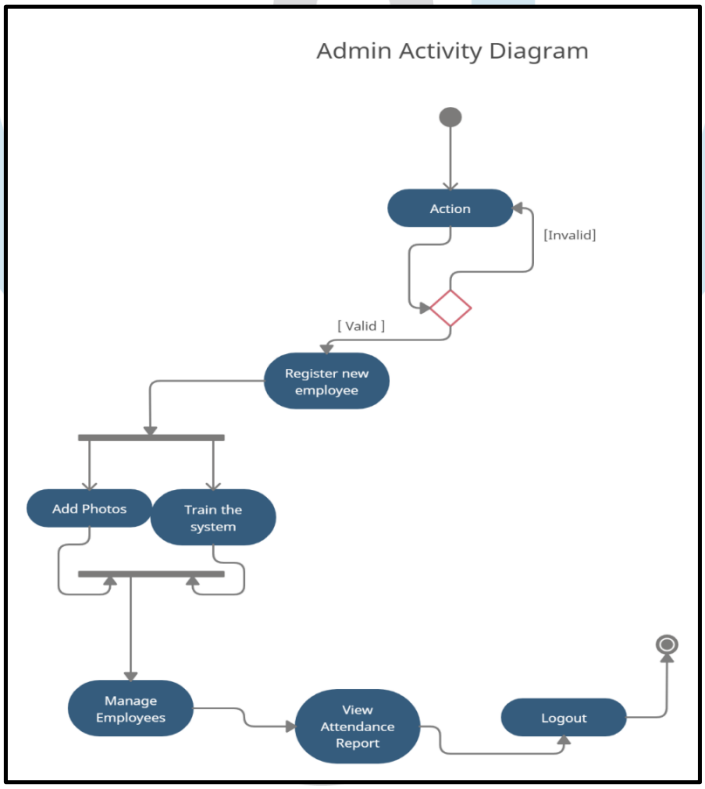
❖ We have 2 types of users of the system.
1. Admin
2. Employee
❖ Following functionalities can be performed by the Admin:
❖ Login
❖ Register new employees to the system
❖ Add employee photos to the training data set
❖ Train the model
❖ View attendance report of all employees. Attendance can be filtered by date or employee.
❖ Following functionalities can be performed by the **Employee**:
❖ Login
❖ Mark his/her time-in and time-out by scanning their face
❖ View attendance report of self

VI. FLOWCHAR



VII. ACTIVITY DIAGRAMS





VIII. Simulation results:

The Attendance management is the significant process that were carry out in every institute to monitor the performance of the student. Every institute does this is its own way. Some of there institute use the old paper or file-based system and some have adopted strategies of automated attendance system using some biometric technique.

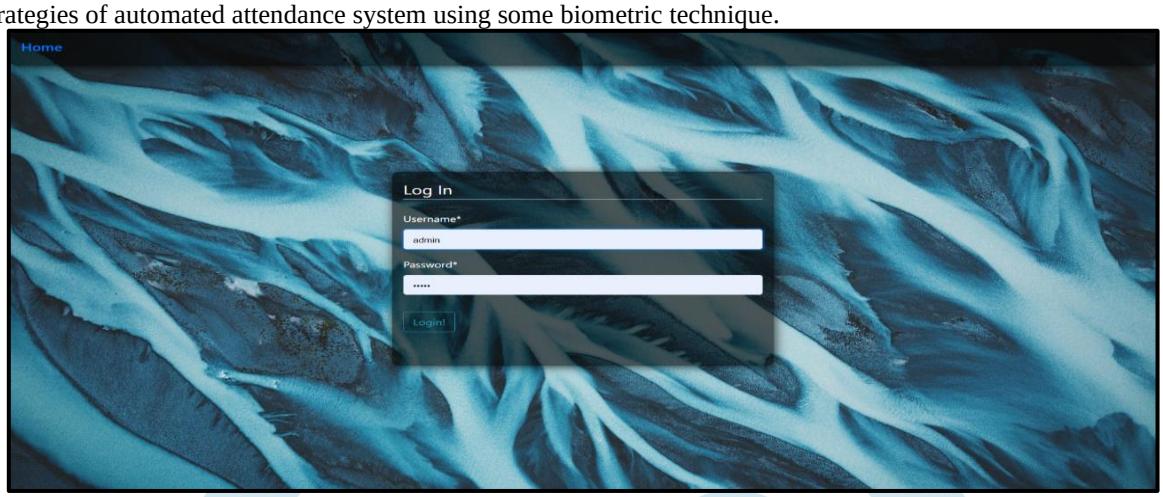
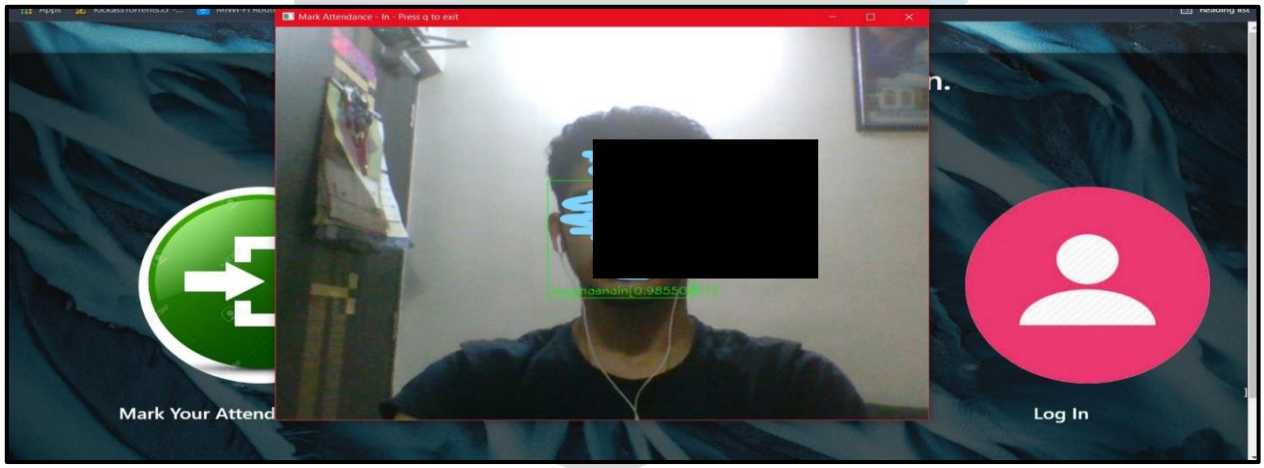


Figure 1.

Figure 2



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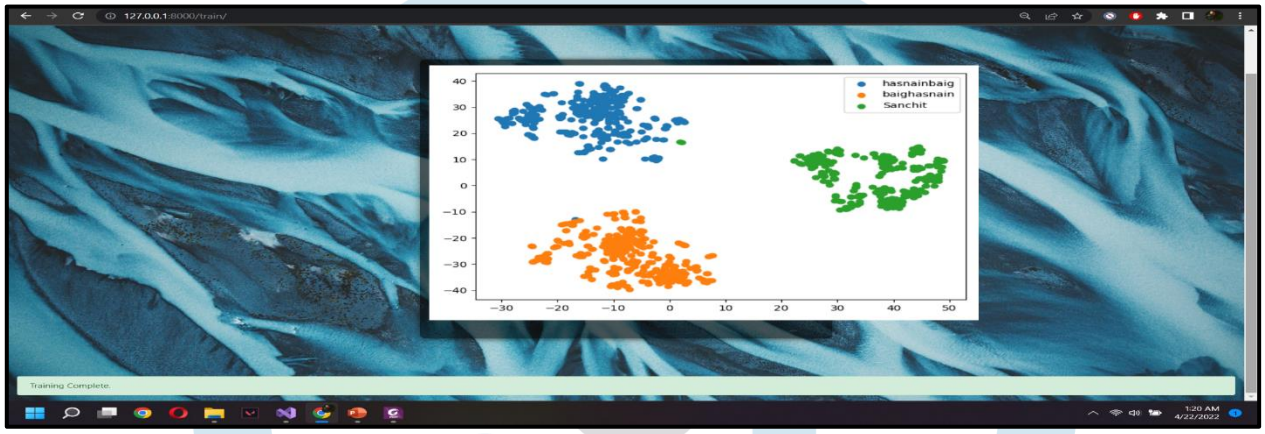

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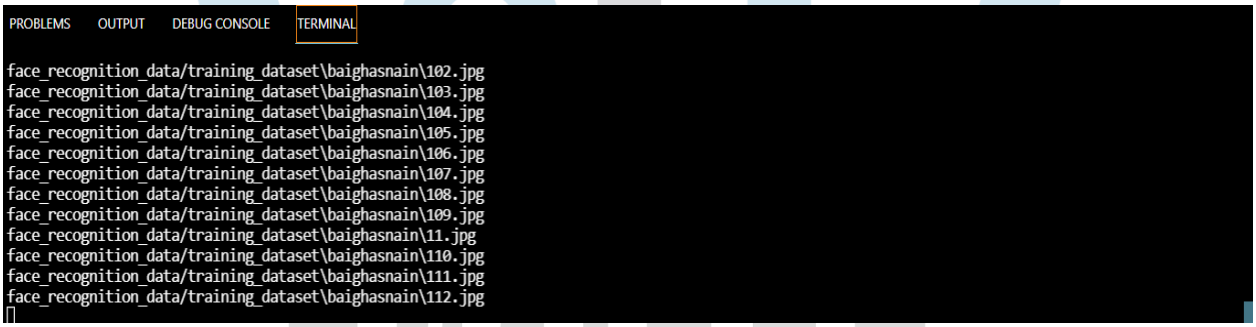

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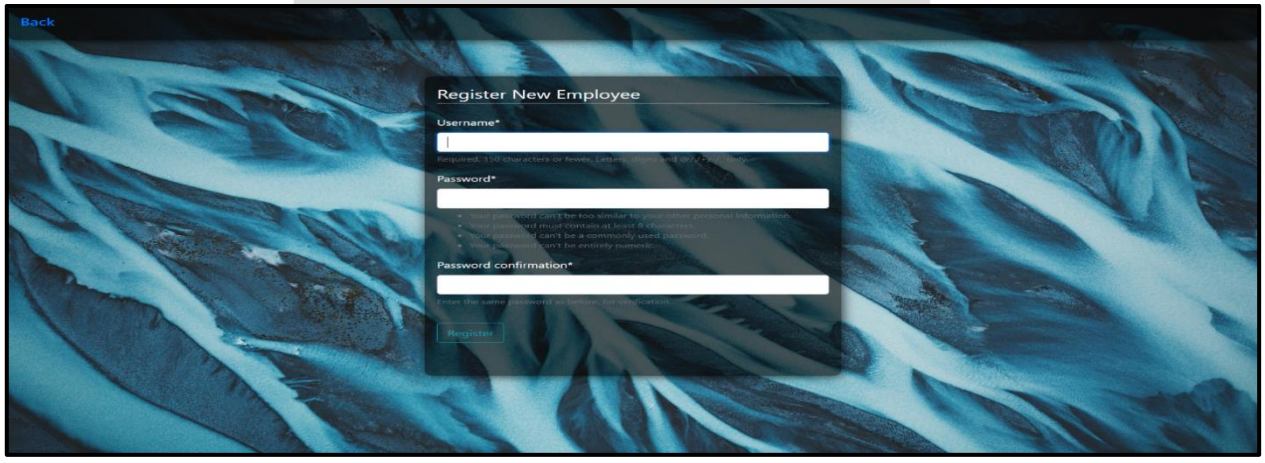

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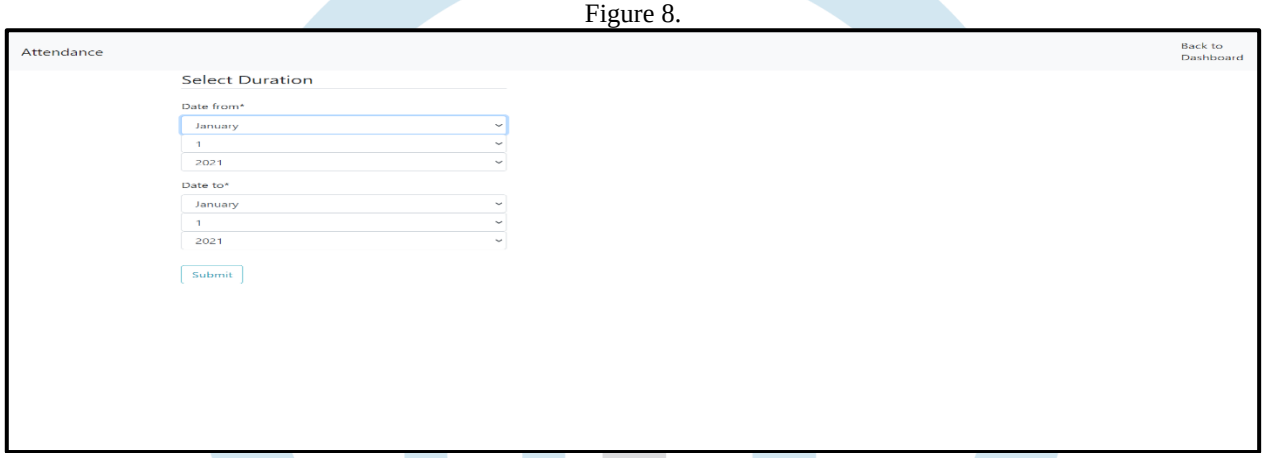

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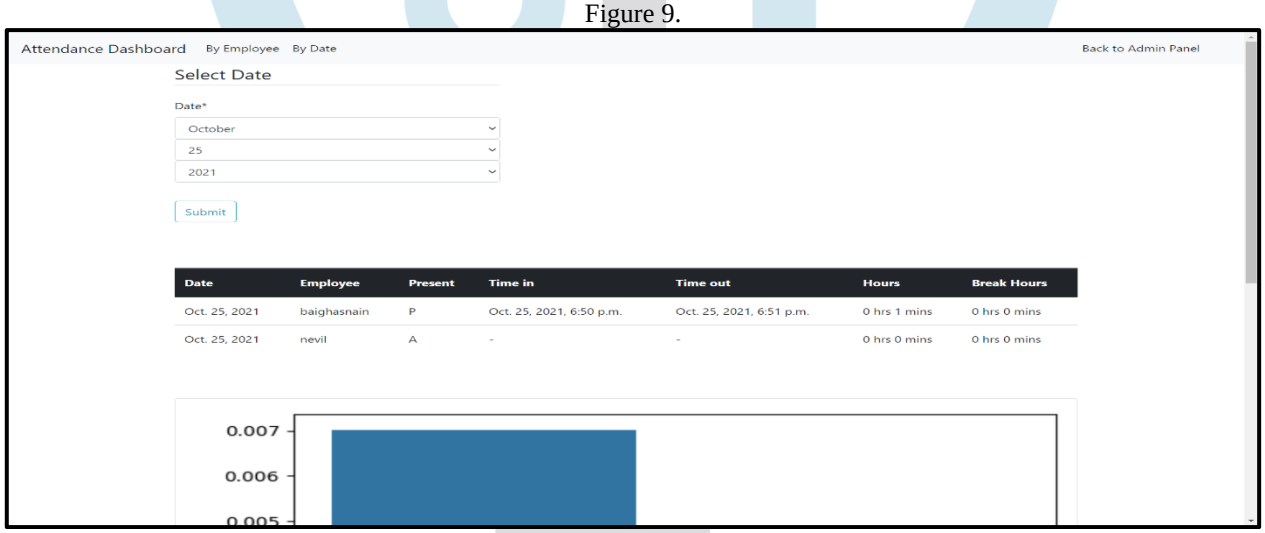

Figure 10.

CONCLUSION

Face Recognition technologies have been associated generally with very costly top secure applications. Certain applications of face recognition technology are now cost effective, reliable and highly accurate. As a result there are no technological or financial barriers for stepping from the pilot project to widespread deployment. This system saves time, reduce the amount of work the administration has to do and will replace the stationary materials with electronic apparatus. Hence a system with the expected results will be developed and there is still room for improvement.

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